The Row Reduction Algorithm

1. Identify the pivot column.
2. Choose a nonzero entry in the pivot column and (if necessary) move it to the pivot position.
3. Create zeros below the pivot using elementary row operations.
4. Repeat steps 1-3 for the sub-matrices obtained by ignoring the column corresponding to the previous pivot position and all the rows above it (if any).
5. Beginning with the rightmost pivot, create zeros above the pivots and make the pivot 1 (if necessary).

Note that step 1–4 will produce a REM and step 5 will convert it to a RREM. see the next example.

Steps 1–4 : forward phase to obtain the REM.

Step 5 : Backward phase to obtain the RREM.

Problem

Solve the system

$$\begin{array}{ccccccc} x & + & y & + & 2z & = & -1 \\ y & + & 2x & + & 3z & = & 0 \\ z & - & 2y & & & = & 2 \end{array}$$

Pivots are hilighted in green.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 2 & 1 & 3 & 0 \\ 0 & -2 & 1 & 2 \end{array}\right] \xrightarrow{-2r_1+r_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -1 & -1 & 2 \\ 0 & -2 & 1 & 2 \end{array}\right] \xrightarrow{-2r_2+r_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -1 & -1 & 2 \\ 0 & 0 & 3 & -2 \end{array}\right]$$

Row-Echelon

$$\xrightarrow{\frac{1}{3}r_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -1 & -1 & 2 \\ 0 & 0 & 1 & -\frac{2}{3} \end{array}\right] \xrightarrow[-2r_3+r_1]{r_3+r_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & \frac{1}{3} \\ 0 & -1 & 0 & \frac{4}{3} \\ 0 & 0 & 1 & -\frac{2}{3} \end{array}\right] \xrightarrow{-r_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & \frac{1}{3} \\ 0 & 1 & 0 & -\frac{4}{3} \\ 0 & 0 & 1 & -\frac{2}{3} \end{array}\right]$$

$$\xrightarrow{-r_2+r_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{5}{3} \\ 0 & 1 & 0 & -\frac{4}{3} \\ 0 & 0 & 1 & -\frac{2}{3} \end{array}\right] \longrightarrow \begin{cases} x_1 = \frac{5}{3} \\ x_2 = -\frac{4}{3} \\ x_3 = -\frac{2}{3} \end{cases}$$

Reduced Row-Echelon

unique solution

Problem

Solve the system

$$\begin{array}{rcrcrcr} -3x_1 & - & 9x_2 & + & x_3 & = & -9 \\ 2x_1 & + & 6x_2 & - & x_3 & = & 6 \\ x_1 & + & 3x_2 & - & x_3 & = & 2 \end{array}$$

$$\left[\begin{array}{ccc|c} -3 & -9 & 1 & -9 \\ 2 & 6 & -1 & 6 \\ 1 & 3 & -1 & 2 \end{array}\right] \xrightarrow{r_3 \leftrightarrow r_1} \left[\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 2 & 6 & -1 & 6 \\ -3 & -9 & 1 & -9 \end{array}\right] \xrightarrow[3r_1 + r_3]{-2r_1 + r_2} \left[\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -2 & -3 \end{array}\right]$$

$$\xrightarrow{2r_2 + r_3} \left[\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right] \longrightarrow 0x_1 + 0x_2 + 0x_3 = 1 \quad \text{or} \quad 0 = 1 \text{ which is impossible}$$

and hence the system is inconsistent (no solution).

The Row Reduction Algorithm

Problem

Solve the system

$$\begin{array}{rcrcrcl} 2x & + & y & + & 3z & = & 1 \\ 2y & - & z & + & x & = & 0 \\ 10z & & & - & 6y & = & 2 \end{array}$$

$$\left[\begin{array}{ccc|c} 2 & 1 & 3 & 1 \\ 1 & 2 & -1 & 0 \\ 0 & -6 & 10 & 2 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 1 & 3 & 1 \\ 0 & -6 & 10 & 2 \end{array}\right] \xrightarrow{-2r_1+r_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -3 & 5 & 1 \\ 0 & -6 & 10 & 2 \end{array}\right]$$

$$\xrightarrow{-2r_2+r_3} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -3 & 5 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-2r_2+r_3} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & -5/3 & -1/3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{-2r_2+r_1} \left[\begin{array}{ccc|c} 1 & 0 & 7/3 & 2/3 \\ 0 & 1 & -5/3 & -1/3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x and y are basic variables, whereas z is a free variable.

$$\left.\begin{array}{l} x + \frac{7}{3}s = \frac{2}{3} \\ x - \frac{5}{3}s = -\frac{1}{3} \\ z = s \end{array}\right\} \longrightarrow \boxed{\begin{array}{l} x = \frac{2}{3} - \frac{7}{3}s \\ y = -\frac{1}{3} + \frac{5}{3}s \\ z = s \\ \text{where } s \in \mathbb{R}. \end{array}}$$